

HALIFAX HARBOR BUOY 44258, NS, Canada

WMO: 995390

Lat: 44.500N

Lon: 63.400W

Elev: 10

StdP: 14.69

Time Zone: -4.00 (NAA)

Period: 02-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.7	13.3	N/A	N/A	N/A	N/A	N/A	N/A	39.1	34.0	36.9	33.2	21.1	300	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	68.5	N/A	66.9	N/A	65.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 34.2	30.7	27.5	DB	4.3	72.3	4.2	3.5	1.3	74.8	-1.2	76.8	-3.5	78.8	-6.6	81.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	29.9	27.9	32.0	38.3	44.6	52.4	59.4	61.6	59.4	53.0	44.1	35.2
	DBStd	12.82	8.19	7.09	5.65	3.71	3.36	3.65	3.25	3.67	3.41	5.06	6.13	7.58
	HDD50	3014	623	617	557	351	172	16	0	0	0	31	189	458
	HDD65	7340	1088	1037	1022	801	633	378	176	114	169	373	626	923
	CDD50	1162	0	0	0	0	4	88	290	361	283	123	13	0
	CDD65	13	0	0	0	0	0	0	1	10	2	0	0	0
	CDH74	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	13.8	17.9	17.0	16.0	13.2	11.1	10.5	9.5	9.8	12.3	15.8	16.3	17.1
Precipitation	PrecAvg	55.5	5.2	4.3	5.1	4.4	4.4	3.9	3.6	3.0	4.1	5.7	5.7	6.0
	PrecMax	62.5	8.3	7.2	9.0	7.9	9.3	9.0	8.4	5.2	12.8	10.3	9.4	9.6
	PrecMin	43.7	1.8	1.1	1.8	1.3	1.8	1.3	0.7	1.5	1.1	1.5	2.7	3.0
	PrecStd	5.9	1.6	1.6	2.1	1.3	1.9	1.9	1.9	1.0	2.3	2.2	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	43.8	45.7	49.2	56.0	64.1	69.0	72.2	71.6	65.0	58.0	53.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	44.5	40.5	42.5	46.3	52.7	60.3	65.9	69.2	66.5	62.6	55.8	50.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	42.9	38.3	40.6	44.1	50.6	58.7	64.7	67.9	65.3	61.0	54.0	48.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	40.9	36.7	38.7	42.7	48.9	57.0	63.6	66.7	64.2	59.6	52.5	46.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	10.4	9.2	8.0	5.6	5.0	5.1	4.9	5.4	5.8	6.3	7.3	10.2
		MCDBR	11.6	10.2	9.2	7.4	7.8	8.1	6.5	5.8	5.7	6.5	8.2	11.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Clear-Sky Solar Irradiance	taub	0.304	0.300	0.326	0.351	0.372	0.402	0.419	0.398	0.356	0.336	0.319	0.299	
	taud	2.483	2.520	2.475	2.448	2.437	2.385	2.362	2.417	2.507	2.555	2.549	2.535	
	Ebn at Noon	261	285	290	289	285	275	269	271	276	268	253	252	
	Edh at Noon	22	25	30	34	35	37	38	35	29	24	21	19	
All-Sky Solar Radiation	RadAvg	455	744	1127	1376	1575	1710	1688	1584	1300	845	488	364	
	RadStd	34	54	77	110	204	161	123	109	83	87	52	28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)		N/A	N/A	+2.09	N/A	N/A	+0.81	N/A	N/A	+99	N/A

Nomenclature: See separate page